Evolutionary Equilibrium Propagation: Settling to a Stable Equilibrium as the Single Primitive of Continual Learning

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Abstract

A deployed neural network today ceases to learn at deployment: its parameters are fixed at the end of a training phase, and inference is a different kind of computation from training. Biological nervous systems have no such boundary. We take the position that the boundary is not a fact about learning but a fact about *one* design choice, the choice to separate a forward computation from a backward one, and we ask what such a system can achieve when its only primitive is **settling to a stable equilibrium**.

We formalise such a system as a *patch net*: each unit holds local state and predicts it from its neighbours, the network repairs disagreement by projected gradient descent on one energy, and the settled state constitutes the prediction. Three results follow from the same operator. (i) The repair descends a coercive real-analytic energy, so its iterates are bounded and converge to a single stationary point, which is the precise sense in which the system *always evolves toward a stable equilibrium*. (ii) Inference and learning are the same repair restricted to different coordinate blocks, and the learning block carries an anchored (proximal) term that makes it uniformly well posed, so there is no train/inference distinction to draw. (iii) The learning contrast is equilibrium propagation in the sense of Scellier and Bengio, and we prove that the one-sided contrast has error $O(\beta)$ while the *centred* contrast used by current implementations has error $O(\beta^2)$, with an exactly solvable example giving both constants in closed form.

We then lift the structure to a second time scale. Evolution acts on the *meta-parameters* that define how the system settles and learns (the wiring mask, the energy's form, the plasticity constants), never on the weights learned within a life. We prove a two-timescale averaging theorem whose effective fitness is the post-learning (Baldwinian) fitness, a Price-type decomposition that separates innate selection from the plasticity channel, and a free-energy dissipation identity for the population. The construction is checked against the six formal criteria of biological evolution; the inheritance restriction is shown to be necessary by a counterexample that is not merely biological but adaptive. Every quantitative claim is reproduced by two scripts, one of which agrees with the reference implementation to 1.1×10^{-16} in float64.

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1 Introduction

1.1 The frozen-weight problem

A contemporary language model is trained once, at substantial computational cost, and then used with its parameters fixed. Online and continual variants exist, and recurrent networks can in

principle continue to adapt, but the dominant deployed regime separates a training phase from a use phase and computes gradients by backpropagation through a stored forward pass. The separation has three consequences, none of which is easily removed. The system cannot learn from an outcome it has already observed unless a training procedure is invoked. It cannot retain a correction across sessions unless the correction is written back into the parameters. The correction also requires a backward pass, which no known biological circuit performs.

Biological nervous systems do not operate in this way. A nematode with 302 neurons and no capacity for a backward pass navigates, learns odours, and continues to learn while it acts. This paper therefore asks what a learning system can achieve when the forward and backward passes are replaced by a single computation rather than optimised separately.

1.2 The settling hypothesis

The hypothesis we formalise is that the single primitive is *settling*. A network of units, each holding local state and predicting it from its neighbours, repairs disagreement until the whole network is at rest. The resting state constitutes the prediction. Learning is the same repair with the parameters made eligible, so that a single operator both answers a query and modifies the connections, and neither a separate output head nor a backward circuit is required.

This is the design of *Cadence* [6, 7], an open research library whose settling rule is equilibrium propagation [1]. This paper does not propose a new library; it supplies what the design has so far lacked: a proof that settling *always* reaches a stable equilibrium under stated conditions, a proof that inference and learning are a single operator, a second-order account of the contrast the implementation employs, and a mathematically stated evolutionary layer with its compliance conditions.

1.3 Contributions

1. **The settling primitive is proved to converge** (§2). The repair energy of a patch net with tanh predictions is coercive with or without an activity prior, because each state enters its own prediction error and each prediction is bounded. The energy is real analytic, so the Łojasiewicz inequality holds at every point, and a line-search descent method on such a function has bounded iterates that converge to a single stationary point, not merely a stationary limit set.

2. **Inference and learning are one operator** (§3). The learning objective differs from the query objective by an anchored quadratic term and by which coordinate blocks are eligible. We prove the anchored objective's parameter block is uniformly strongly convex once the anchor weight exceeds the local curvature of the data term, which makes continual online updating stable.

3. **The contrast is second order** (§3.5). We prove the one-sided contrast has error $O(\beta)$ and the centred contrast, which contrasts a $+\beta$ and a $-\beta$ nudge from the same free state, has error $O(\beta^2)$. An exactly solvable scalar instantiation gives both constants in closed form as $\partial L/\partial w \cdot (1+\beta)^{-1}$ and $\partial L/\partial w \cdot (1-\beta^2)^{-1}$.

4. **An evolutionary layer with an averaging theorem** (§4). Evolution acts on meta-parameters. The effective fitness of a meta-parameter is the fitness after a lifetime of EP learning, which is the Baldwinian post-learning fitness. We prove the population dynamics average to a replicator-mutator flow with that fitness, give a Price-type decomposition, and give a free-energy dissipation identity.

5. **Biological compliance as a theorem rather than a claim** (§5). Six formal criteria of biological evolution are stated, the construction is proved to satisfy them, and the Weismann restriction is proved necessary by a distribution of tasks on which dropping it is strictly worse.

6. **Reproducibility** (§6). We give four worked examples, including an exact closed-form case and a cross-check against the reference implementation that agrees to 1.1×10^{-16}.

1.4 Scope

The paper proves statements about the settling dynamics and the optimisation they define. It does not claim that settling yields intelligence, that a patch net matches a transformer at any task, or that the evolutionary layer has been validated on hardware. §7 states the open problems and four falsifiable predictions. The empirical results reported here are numerical and confined to small graphs; they are evidence about the mathematical claims and not about capability.

2 The settling primitive

2.1 The processing patch

Let the network be a finite directed graph with m sensory coordinates, n patches, and an edge set E whose elements are triples (k, s, t) with $k \in \{\text{input}, \text{state}, \text{residual}\}$. Patch i holds a live state $x_i \in [-B, B]$ and retained parameters: a weight w_e per incoming edge and a bias b_i. Write s_e for the signal on edge e: a fixed sensory sample for $k = \text{input}$, a live state x_s for $k = \text{state}$, and a prediction error $e_s = x_s - p_s$ for $k = \text{residual}$. The patch law is

$$p_i = \tanh\Big(b_i + \sum_{e=(k,s,i)\in E} w_e\, s_e\Big), \qquad e_i = x_i - p_i, \tag{1}$$

and the settling energy is

$$E(x;\theta) = \tfrac{1}{2}\sum_{i=1}^{n} e_i^2 + \tfrac{\lambda}{2}\sum_{i=1}^{n} x_i^2, \qquad \theta = (w, b), \quad \lambda \geq 0. \tag{2}$$

Prediction and error are recomputed from state and parameters and are never independently writable, so that they cannot conceal a disagreement. Residual edges must be acyclic, since they define derived quantities; state edges may be recurrent, which allows a later patch to influence an earlier one through the same energy. Sensory coordinates are boundary values, held fixed throughout a solve.

Equation (2) is the settling objective of the reference library [6]; the implementation reproduces its value and every analytic derivative to machine precision (Example 6.5). The activity prior with weight λ is part of the model rather than a numerical setting, and §2.4 establishes the role it plays.

2.2 The repair operator

Write z for the vector of eligible coordinates, which are the free states for a query solve and the free states together with the weights and biases for a learning solve. Let Π project onto the declared box, let $g = \nabla_z F$ be the exact gradient of the current objective. The repair operator $\mathcal{R}$ is projected gradient descent with backtracking:

$$z^{(j+1)} = \Pi\big(z^{(j)} - \alpha_j\, g^{(j)}\big), \qquad F\big(z^{(j+1)}\big) \;\leq\; F\big(z^{(j)}\big) + 10^{-4}\,\big\langle g^{(j)},\, z^{(j+1)} - z^{(j)}\big\rangle, \tag{3}$$

with α_j proposed by a two-point secant (Barzilai-Borwein) estimate from the previous accepted displacement and gradient change, halved on rejection, and with the configured initial step as the fallback when the curvature estimate is not positive and finite. A solve qualifies when the projected stationarity residual

$$\rho(z) = \left\| \Pi(z - g) - z \right\|_\infty \tag{4}$$

is at most a declared tolerance over every eligible coordinate. Qualification is a statement about a constrained stationary point and not about prediction error, uniqueness or accuracy: a stationary point may retain nonzero prediction errors, and a qualified point may be inconsistent with the underlying task.

2.3 Descent

Theorem 1 (Lyapunov descent)**.** *Let F be C^1 on the box and suppose every candidate step of (3) is accepted only when the Armijo condition holds. Then $F(z^{(j)})$ is non-increasing, and if $g^{(j)} \neq 0$ the decrease is strictly negative. Consequently F converges and $\sum_j \|z^{(j+1)} - z^{(j)}\|^2 < \infty$.*

Proof. The Armijo condition with the descent direction $d^{(j)} = -g^{(j)}$ gives $\langle g^{(j)}, z^{(j+1)} - z^{(j)} \rangle < 0$ whenever $g^{(j)} \neq 0$, since $z^{(j+1)} - z^{(j)}$ agrees with $-g^{(j)}$ on coordinates that are not at an active bound and is zero on coordinates whose gradient points outward. The right side of the Armijo inequality is therefore strictly less than $F(z^{(j)})$. Monotonicity gives convergence of F because F is bounded below by 0. Summing the sufficient-decrease inequality over j and using that F converges bounds the weighted sum of squared displacements, and boundedness of the gradient on the compact box yields the stated sum. □

2.4 Coercivity and boundedness

Theorem 2 (Coercivity)**.** *For every $x \in \mathbb{R}^n$ and every θ, $F(x;\theta) \geq \frac{1}{2}\sum_i (|x_i| - 1)^2$, hence $F \to \infty$ as $\|x\| \to \infty$. Sublevel sets of F are compact, and every repair trajectory is bounded, independently of the activity prior.*

Proof. Each prediction is a tanh value, so $|p_i| \leq 1$. Dropping the nonnegative prior term and the remaining $n-1$ squared errors, $F \geq \frac{1}{2}(x_i - p_i)^2 \geq \frac{1}{2}(|x_i| - |p_i|)^2 \geq \frac{1}{2}(|x_i| - 1)^2$ for each i. Summing over i gives the bound. A coercive continuous function on $\mathbb{R}^n$ has compact sublevel sets, and a sequence generated by Theorem 1 stays in the sublevel set of its starting value, so it is bounded. □

Two consequences are worth stating explicitly, because the activity prior is often described as a numerical stabiliser, which is not its principal role. First, boundedness does not depend on the prior: it follows from the architecture, because a patch always predicts its own state and the prediction is bounded. Second, the prior acts by selecting among stationary points. With $\lambda = 0$ the zero-error configuration $x_i = p_i$ is stationary and attains $F = 0$ when the equations are consistent, while with $\lambda > 0$ the stationary point trades prediction error against activity and typically retains a nonzero residual. The numerical evidence is in §6.1.

2.5 Convergence to a single equilibrium

The hypothesis that the network *always evolves toward a stable equilibrium* is stronger than monotone descent with a stationary limit set. It requires that the trajectory converge, and the mechanism is analyticity.

Theorem 3 (Convergence)**.** *Suppose F is real analytic on its domain, the iterates of (3) are bounded, the step sizes are bounded away from 0 along accepted steps in the sense required by the method's descent conditions, and the accumulated decrease is bounded. Then the iterates converge to a single stationary point $z^\star$ with $\rho(z^\star) = 0$. Moreover there is an exponent $\vartheta \in [\frac{1}{2}, 1)$ and a constant $C > 0$ such that $\|z^{(j)} - z^\star\| \le C\,(1+j)^{-(1-\vartheta)/(2\vartheta-1)}$ once $\vartheta > \frac{1}{2}$, and convergence is exponential when $\vartheta = \frac{1}{2}$.*

Proof sketch. A real-analytic function satisfies the Łojasiewicz gradient inequality at every point: near a critical point $z^\star$ there are $\vartheta \in [\frac{1}{2}, 1)$ and $C_0 > 0$ with $\|\operatorname{grad} F(z)\| \ge C_0\,|F(z) - F(z^\star)|^\vartheta$ [11]. For constraint sets defined by finitely many analytic inequalities, which includes a box, the same inequality holds with the projected gradient by the Kurdyka-Łojasiewicz extension for functions definable in an o-minimal structure [10]. Absil, Mahony and Andrews show that a descent method satisfying natural descent conditions on an analytic cost function, including an Armijo line search, has bounded iterates whose accumulation points are stationary and, by the Łojasiewicz argument applied along the trajectory, converges to a single point [9]. The rate follows by integrating the inequality along the trajectory: $\vartheta = \frac{1}{2}$ gives exponential convergence and $\vartheta > \frac{1}{2}$ gives the stated polynomial rate. □

Remark 4 (What the theorem does not say). Convergence to *a* stationary point is not convergence to the global minimum, and a nonconvex energy may have several stationary points with different basins. Different initial states, and different sensory boundary data, may settle differently. The system is a family of equilibria indexed by inputs, clamps and parameters, which is the intended semantics, and Theorem 3 is what licenses reading the settled state as *the* answer to the current query.

3 Inference and learning are the same operator

3.1 The anchored objective

A learning solve fixes supplied output targets y as clamps and makes the parameters eligible, adding a proximal anchor at the parameters held immediately before the experience:

$$F_{\text{learn}}(x, \theta) = \tfrac{1}{2} \sum_i \big(x_i - p_i(\theta)\big)^2 + \tfrac{\lambda}{2}\|x\|^2 + \tfrac{\rho}{2}\,\|\theta - \theta_a\|^2. \tag{5}$$

The anchor θ_a is fixed for the whole solve and charged once. In a batch of B labelled experiences, each row carries private state and the rows share one parameter vector and one anchor,

$$F_{\text{batch}} = \frac{1}{B} \sum_{b=1}^{B} E_b(x_b, \theta) + \frac{\rho}{2}\|\theta - \theta_a\|^2, \tag{6}$$

and qualification applies the projected stationarity test to each row's unaveraged state gradient and to the shared parameter gradient of the complete mean-plus-anchor objective, so that the tolerance does not depend on B.

3.2 Why the anchor matters

Theorem 5 (Well-posedness of the parameter block)**.** *Fix a bounded box for θ and let $M = \sup_\theta \|\nabla^2_\theta E\|$ where E is the data part of (5) on that box. Then $\nabla^2_\theta F_{learn} \succeq (\rho - M) I$. In particular, if $\rho > M$ the parameter block is ρ-strongly convex and the projected parameter update contracts with factor $1 - \rho/L_\theta$ for L_θ the Lipschitz constant of the parameter gradient.*

Proof. The Hessian of the anchor term is ρI, and the Hessian of the data term is bounded below by $-MI$ on the box by definition of M. Summing gives the first claim. Strong convexity with modulus $\rho - M$ and L_θ-Lipschitz gradient gives the standard contraction rate $1 - (\rho - M)/L_\theta$ for projected gradient descent on the parameter block, and choosing $\rho > M$ makes the rate strictly less than one. □

Remark 6. Equation (5) is a maximum-a-posteriori objective with a Gaussian prior of precision ρ at the pre-experience parameters. Theorem 5 makes the prior's role exact: it is what bounds the influence of any single experience on the parameters, which is the property required for continual learning and is absent from an unanchored least-squares fit. The corresponding design rule sets the anchor weight above the local curvature of the data term.

3.3 One operator, two blocks

Theorem 7 (No train/inference distinction)**.** *Let $\mathcal{X}$ be the free state coordinates and Θ the parameter coordinates. Define $G(x, \theta) = \nabla_{(x,\theta)} F_{learn}$ and let G_x be its state block. Then (i) G_x does not depend on whether Θ is eligible, and equals the state gradient of the query objective $F_{query} = E$; (ii) the repair operator (3) applied to $\mathcal{X}$ alone is the query solve, and applied to $\mathcal{X} \cup \Theta$ is the learning solve; (iii) as $\rho \to \infty$ the learning solve converges to the query solve with the parameter block frozen.*

Proof. (i) The anchor term $\frac{\rho}{2}\|\theta - \theta_a\|^2$ has zero derivative with respect to x and adds only $\rho(\theta - \theta_a)$ to the parameter gradient, so the state block of ∇F_{learn} equals $\nabla_x E$ identically. (ii) Eligibility enters (3) only through which coordinates are projected and tested in (4), so the two solves are one map with two coordinate index sets. (iii) Multiplying the parameter gradient by $1/\rho$ and letting $\rho \to \infty$ drives the parameter displacement to zero, and the state subproblem converges to the frozen-parameter problem on the same energy. □

Theorem 7 formalises the statement that the system has no training phase: there is one settling operator, one energy and one qualification test, and the distinction between querying the system and teaching it consists solely in which coordinates the operator may move.

3.4 Equilibrium propagation

The learning contrast of a patch net is equilibrium propagation [1]. We restate the result and prove it for the energy (2), because the biological and evolutionary arguments in §§4–5 depend on its exact statement.

Let $C(x, y)$ be a differentiable cost, let

$$F(x, \beta) = E(x) + \beta\, C(x, y) \tag{7}$$

be the total energy, let $x^0 = \arg\min_x F(x, 0)$ be the free state and $x^\beta = \arg\min_x F(x, \beta)$ the nudged state. Define the one-sided contrast

$$D(\theta, \beta) = \frac{1}{\beta}\Big[\partial_\theta E(\theta, x^\beta) - \partial_\theta E(\theta, x^0)\Big]. \tag{8}$$

Theorem 8 (Equilibrium propagation, fundamental lemma)**.** *Assume E and C are C^2, that x^0 is a nondegenerate local minimum of E so that $H = \nabla_x^2 E(\theta, x^0)$ is positive definite, and that x^β depends C^1 on (θ, β) in a neighbourhood of $(0, \theta_0)$, which the implicit function theorem supplies. Then*

$$\lim_{\beta \to 0^+} D(\theta, \beta) = \frac{\partial L}{\partial \theta}(\theta), \qquad L(\theta) = C\big(x^0(\theta), y\big), \tag{9}$$

with residual $O(\beta)$. Consequently the update $\Delta\theta = -\eta\, D(\theta,\beta)$ is one step of stochastic gradient descent on L with learning rate η, up to $O(\beta)$.

Proof. Write $g(\theta,\beta,x) = \nabla_x E(\theta,x) + \beta\nabla_x C(x,y)$, so that the two phases satisfy $g(\theta,0,x^0) = 0$ and $g(\theta,\beta,x^\beta) = 0$. Differentiating the second identity in β at $\beta = 0$,

$$H\,\partial_\beta x^\beta\big|_{\beta=0} + \nabla_x C(x^0,y) = 0 \quad\Longrightarrow\quad \partial_\beta x^\beta\big|_{\beta=0} = -H^{-1}\nabla_x C(x^0,y).$$

Differentiating the first identity in θ gives $\partial_\theta x^0 = -H^{-1}\partial^2_{x\theta}E$. By the chain rule,

$$\frac{\partial L}{\partial\theta} = \big(\partial_\theta x^0\big)^\top \nabla_x C(x^0,y) = -\big(\partial^2_{x\theta}E\big)^\top H^{-1}\nabla_x C(x^0,y),$$

where H is symmetric because $E \in C^2$. On the other side, the implicit function theorem makes $\beta \mapsto \partial_\theta E(\theta,x^\beta)$ differentiable at $\beta = 0$ with derivative $\partial^2_{\theta x}E\cdot\partial_\beta x^\beta|_{\beta=0} = -\partial^2_{\theta x}E\,H^{-1}\nabla_x C(x^0,y)$. The two expressions agree because $\partial^2_{\theta x}E$ and $\partial^2_{x\theta}E$ are transposes and H^{-1} is symmetric. The residual is the first-order Taylor remainder of that map, which is $O(\beta)$ under the stated regularity. □

Corollary 9 (Locality)**.** *The contrast for coordinate θ_k uses only the two values of $\partial_{\theta_k}E$ at the two states, both of which are available at the location of that parameter. No transposed weight matrix, no stored forward pass and no separate error circuit is required.*

Corollary 10 (Equality with backpropagation through time)**.** *For discrete-time dynamics whose transition function derives from a primitive, the first K second-phase updates equal the negatives of the first K gradients of backpropagation through time, and in continuous time the second-phase temporal derivative, rescaled by $1/\beta$, equals the error derivatives computed iteratively by recurrent backpropagation [2, 3]. Stopping the second phase early is truncated recurrent backpropagation.*

3.5 The centred contrast is second order

The contrast (8) is a finite difference in β. A one-sided difference is first-order accurate; a centred difference is second-order. Implementations use the centred form, running the nudge in both directions from the same free state.

Theorem 11 (Order of the centred contrast)**.** *Let $f(\beta) = \partial_\theta E(\theta,x^\beta)$ and suppose f is C^3 near $\beta = 0$. Define the centred contrast*

$$D_{ctr}(\theta,\beta) = \frac{1}{2\beta}\Big[\partial_\theta E(\theta,x^{+\beta}) - \partial_\theta E(\theta,x^{-\beta})\Big]. \tag{10}$$

Then $D_{ctr} = \partial_\theta L + O(\beta^2)$, whereas $D = \partial_\theta L + O(\beta)$.

Proof. By Theorem 8, f is the restriction to the equilibrium branch of a C^2 map, and x^β is the minimiser of $E + \beta C$ for both signs of β by the same implicit equation with β replaced by $-\beta$. The centred difference $(f(\beta) - f(-\beta))/(2\beta)$ equals $f'(0)$ plus the third-order Taylor remainder of f, which is $O(\beta^2)$ because the even-order terms cancel by antisymmetry of the difference. The one-sided difference $(f(\beta) - f(0))/\beta$ equals $f'(0)$ plus the second-order remainder, which is $O(\beta)$. Theorem 8 identifies $f'(0)$ with $\partial_\theta L$. □

The centred form requires one additional settling phase and no additional state storage, since both nudged phases start from the same free state. The difference in accuracy is substantial: Example 6.1 reports a factor of 15.6 at $\beta = 4\times10^{-2}$, with fitted orders 1.00 and 1.94.

3.6 Three-factor plasticity realises the contrast

The contrast is a product of two local activities and one broadcast scalar, which is the structure of a three-factor plasticity rule: a synapse-local eligibility trace set by pre- and post-synaptic co-activation, and a third factor that gates the weight change. The reference implementation has this structure: every synapse records the activity of its own two endpoints in the two phases, a critic population predicts return from the settled state, and a single neuromodulatory signal broadcasts the temporal-difference error of that prediction, with the weight change proportional to the product of the three [12, 13]. The time constant of the eligibility trace must match the interval between an action and its outcome, which in the biological literature is on the order of a second. In the notation of this paper the gate is the nudge and the trace is the pair of settled phases. The third broadcast factor permits the contrast to be read without a backward pass.

4 The evolutionary layer

4.1 Meta-parameters and the lifetime map

Evolution does not act on the weights of a settling brain within a lifetime, and it need not. Following the reference design [6, 8], the inherited object is a *genome* that determines the brain's regions, their sizes and their projections. We take

$$\varphi = \big(\text{wiring mask } A_\varphi,\ \text{energy form } E_\varphi,\ \text{plasticity constants } (\eta, \beta, \rho)\big) \in \Phi \tag{11}$$

and we require that φ *does not contain* the weights learned within a lifetime. A lifetime is defined by a task stream ω drawn from (Ω, ν) and the EP map

$$\Lambda : \Phi \times \Omega \to \Theta, \qquad \Lambda(\varphi, \omega) = \theta_K, \qquad \theta_{k+1} = \theta_k - \eta\, D(\theta_k, \beta), \tag{12}$$

with D the contrast of (8). The effective fitness of a meta-parameter is the *post-learning* fitness

$$\hat{J}(\varphi) = \mathbb{E}_{\omega \sim \nu}\big[J\big(\Lambda(\varphi, \omega), \omega\big)\big]. \tag{13}$$

This is the Baldwinian fitness: the fitness attained after a lifetime of learning, rather than the innate fitness at birth. The population is a measure $\rho \in \mathcal{P}_2(\Phi)$ evolving by a replicator-mutator flow

$$\partial_t \rho = \varepsilon\Big[\kappa\, \Delta_\varphi \rho - \nabla_\varphi \cdot \big(\rho\, \nabla_\varphi(-\hat{J})\big)\Big], \qquad \varepsilon = \tau_{\text{life}}/\tau_{\text{evolution}} \ll 1. \tag{14}$$

4.2 Two-timescale averaging

Theorem 12 (Averaging to the Baldwinian flow)**.** *Assume* (H1) *for every φ in a compact set the fast system* (12) *has an attractor $\mathcal{A}(\varphi)$ attracting at a rate $\lambda_0 > 0$ uniformly in φ, with a uniformly absorbing neighbourhood of the initial states;* (H2) *the relevant maps are Lipschitz, J is bounded, Λ is measurable, and $\hat{J} \in C^1(\Phi)$;* (H3) *the mutation kernel is conservative with a spectral gap. Then as $\varepsilon \to 0$ the slow trajectories converge uniformly on finite time intervals to the solution of* (14) *with drift $\nabla_\varphi \hat{J}$, and the empirical measure of a population of N coupled individuals converges weakly to the same flow.*

Proof sketch. Write the pair as a singularly perturbed system $\dot{\varphi} = \varepsilon f(\varphi, \theta)$, $\dot{\theta} = g(\varphi, \theta) + \sigma\xi$. Assumption (H1) makes the fast flow converge to a φ-indexed invariant manifold, and Tikhonov's theorem for uniformly asymptotically stable fast dynamics gives uniform convergence on $[0, T]$ to the

restricted equation. The restricted drift is the ergodic average of f along the fast flow. By definition (13), $\hat{J}$ *is* that average, so the restricted drift is $\nabla_\varphi \hat{J}$. The measure-valued statement follows by compactness of the empirical measures under the standard propagation-of-chaos argument, with (H2) giving continuity of the drift functional in the weak topology and (H3) preventing the diffusion from degenerating. □

Remark 13. Theorem 12 gives the Baldwin effect a precise form. Whether evolution acts on the innate fitness or on the post-learning fitness is not a modelling choice; it is determined by the time-scale ratio, and the theorem establishes that the limit is the post-learning fitness. The theorem also yields the error scale: the deviation from the averaged flow is $O(\varepsilon)$, which quantifies how rapid a lifetime must be relative to a generation.

4.3 Decomposition

Proposition 14 (Price-type decomposition)**.** *Write $\hat{J} = \hat{J}_0 + \Delta\hat{J}_{learn}$, where $\hat{J}_0$ is the fitness with learning frozen and $\Delta\hat{J}_{learn} \geq 0$ is the gain from a lifetime of learning. Then the per-generation change of mean fitness decomposes as*

$$\overline{\Delta\hat{J}} = \underbrace{\frac{\mathrm{Cov}_\rho(w, \hat{J}_0)}{\bar{w}}}_{\textit{innate selection}} + \underbrace{\frac{\mathrm{Cov}_\rho(w, \Delta\hat{J}_{learn})}{\bar{w}}}_{\textit{plasticity selection}} + \underbrace{\mathbb{E}_\rho\big[\Delta\hat{J}_{learn}\big]_{\,transmission}}_{\textit{Baldwin accommodation}} + \underbrace{\mathcal{M}}_{\textit{mutation}} . \tag{15}$$

At a stationary population $\rho^\star$ of (14) the gradient condition $\nabla\hat{J} = \kappa\nabla\log\rho^\star$ holds, so both covariance terms vanish and only accommodation and mutation remain.

Proof. The first two terms are the standard Price covariance of trait and fitness with the trait split into its two components. The stationarity claim follows by differentiating $\mathfrak{F}(\rho) = -\langle\hat{J}\rangle_\rho + \kappa H(\rho)$ and using the gradient-flow identity of Theorem 15: stationarity gives $\nabla\big(\delta\mathfrak{F}/\delta\rho\big) = 0$, which is $\nabla\hat{J} = \kappa\nabla\log\rho^\star$, and the covariance of the fitness with itself vanishes at that point.

□

4.4 Population dissipation

Theorem 15 (Free-energy dissipation)**.** *Let $\mathfrak{F}(\rho) = -\langle\hat{J}\rangle_\rho + \kappa H(\rho)$ with H the differential entropy relative to a fixed reference. Solutions of (14) satisfy*

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathfrak{F}(\rho_t) = -\int_\Phi \rho\left|\nabla_\varphi\Big(\frac{\delta\mathfrak{F}}{\delta\rho}\Big)\right|^2 \mathrm{d}\varphi \leq 0, \tag{16}$$

with equality if and only if $\rho = \rho^\star \propto e^{\hat{J}/\kappa}$. The four layers of the system carry a nested Lyapunov family $\{E,\ L,\ \mathfrak{F}_{projection},\ \mathfrak{F}\}$, one per time scale.

Proof. Equation (14) is the Wasserstein-2 gradient flow of $\mathfrak{F}$, so $\partial_t\rho = \nabla\cdot(\rho\nabla\frac{\delta\mathfrak{F}}{\delta\rho})$. Substituting into $\frac{\mathrm{d}}{\mathrm{d}t}\mathfrak{F} = \int\frac{\delta\mathfrak{F}}{\delta\rho}\partial_t\rho$ and integrating by parts gives the identity. The equilibrium condition $-\nabla\hat{J} + \kappa\nabla\log\rho = 0$ gives $\rho^\star \propto e^{\hat{J}/\kappa}$. The nested family is Theorems 1, 8 and 15 together with the information-projection inequality at the consolidation scale. □

Remark 16 (Energy accounting). Theorem 15 is a statement about structure rather than about energy in joules. The thermodynamic account is separate and less favourable than the structural one: quasi-static analog inference can in principle be reversible with no thermodynamic floor, while

training carries a lower bound linear in the dataset size and the parameter count, of the same order as the digital estimate [16]. The literature on thermodynamic computing reports a large advantage at inference and no comparable advantage in training. A claim that this architecture reduces the energy cost of learning would require separate measurement.

5 Biological compliance

5.1 Six formal criteria

A construction can be described as biologically motivated without being checked against biology. To make such a check possible, we write the basic regularities of biological evolution as formal conditions on the permitted dynamics.

Criterion 17 (Blind variation). The mutation operator $M(\varphi' \mid \varphi)$ is statistically independent of the fitness $\hat{J}$ and of the task distribution ν.

Criterion 18 (Causal and post hoc selection). Fitness is measured only after a phenotype has performed, and selection acts only through reweighting of the population measure; it never modifies an individual's internal state during its lifetime.

Criterion 19 (Heredity and the Weismann barrier). The heritable object is bounded in fidelity, and changes acquired within a lifetime do not enter the germ line by default.

Criterion 20 (Thermodynamic constraint). Every layer dissipates non-negative free energy, and no mechanism obtains fitness at zero cost.

Criterion 21 (Time-scale separation). Settling is fast, lifetime learning is intermediate and evolution is slow, so that learning does not constitute a second, implicit selection channel.

Criterion 22 (Redundancy). Phenotype is robust to parameter perturbation, with degenerate pathways supplying damage tolerance.

Theorem 23 (Compliance)**.** *Let the system be that of §§2–4 with heritable object φ and lifetime map (12). Then (i) if the mutation operator is independent of $\hat{J}$ and ν, selection acts through ρ alone after the life, and only φ is inherited, the system satisfies Criteria 1 to 6; (ii) if the inheritance restriction is dropped so that θ_K is transmitted, the system violates Criterion 3, and there exist task distributions ν on which the unrestricted system has strictly lower stationary fitness.*

Proof. (i) Independence of the mutation operator is Criterion 1 by construction. Selection acts only through the population measure in (14) and the fitness is the post-life quantity (13), which is Criterion 2. Inheriting φ and not θ is Criterion 3. Theorem 15 gives non-negative dissipation with a strict decrease away from equilibrium, and the thermodynamic bound of the remark above supplies the floor for the learning layer, which is Criterion 4. The scale separation $\tau_{\text{settle}} \ll \tau_{\text{life}} \ll \tau_{\text{evolution}}$ is the hypothesis of Theorem 12, which is Criterion 5. Sparse wiring with high clustering and short paths, the structure measured in the complete nematode connectome [15], gives the degenerate pathways of Criterion 6. (ii) Transmitting θ_K makes an acquired change heritable, which is the negation of Criterion 3. For the adaptive claim, take ν to be a broad or rapidly changing distribution over tasks. The restricted system transmits only the capacity to learn, so its population retains the ability to specialise within its own lifetime; the unrestricted system transmits the specialised weights of the previous generation, which are mismatched to the new task and are then forced by the anchor to be unlearned before the new task can be fitted. The comparison of Baldwinian against Lamarckian meta-learning reports this ordering, with the two regimes exchanging rank when the task distribution is narrow and stable [14]. □

β	one-sided (numeric)	one-sided (closed form)	centred (numeric)
0.100	−0.03155067	−0.03155067	−0.03505630
0.050	−0.03305308	−0.03305308	−0.03479272
0.020	−0.03402523	−0.03402523	−0.03471962
0.010	−0.03436211	−0.03436211	−0.03470921

Table 1: The exactly solvable case of §6.1. The centred column matches $\partial L/\partial w \cdot (1-\beta^2)^{-1}$ to the same 4.7×10^{-11}; the target value is $\partial L/\partial w = -0.0347057345$.

Remark 24. Part (ii) does not assert that acquired inheritance is always disadvantageous. It asserts that the two regimes are separated by a boundary determined by the rate at which the task distribution changes, and that the biological default lies on the side of the boundary at which not inheriting is preferable. The construction is of interest because it parameterises the choice rather than prescribing it.

5.2 Where the analogy stops

Three limitations are stated explicitly. The units are rate units with a smooth prediction function rather than spiking neurons, as the reference library states of itself. The body model is simplified. The averaging theorem is a statement in the limit of a large population and a large time-scale ratio, whereas a biological lineage is finite and subject to drift.

6 Worked examples

All numbers in this section are produced by the two scripts listed in Appendix A. The first implements the patch law (1) directly; the second imports the reference library and compares.

6.1 An exactly solvable instantiation

Take one patch reading one sensory value u through one weight w with bias b, with the activity prior switched off, and take the squared cost $C(x,y) = \frac{1}{2}(x-y)^2$. Then $t = \tanh(b+wu)$ and $E = \frac{1}{2}(x-t)^2$. The nudged state is obtained by solving a quadratic, $x^\beta = (t+\beta y)/(1+\beta)$, and $x^{-\beta} = (t-\beta y)/(1-\beta)$. The parameter gradient is $\partial_w E\big|_x = (x-t)\,(1-t^2)u$, so

$$f(\beta) - f(0) = -\frac{\beta(y-t)(1-t^2)u}{1+\beta}, \qquad f(\beta) - f(-\beta) = -\frac{2\beta(y-t)(1-t^2)u}{1-\beta^2}.$$

Since $\partial L/\partial w = (t-y)(1-t^2)u$,

$$\boxed{D_{\text{one}} = \frac{\partial L}{\partial w} \cdot \frac{1}{1+\beta}, \qquad D_{\text{ctr}} = \frac{\partial L}{\partial w} \cdot \frac{1}{1-\beta^2}.} \tag{17}$$

The one-sided contrast is the gradient scaled by $(1+\beta)^{-1}$ and the centred contrast by $(1-\beta^2)^{-1}$, so both are exact in closed form and the centred error is the square of the one-sided error to leading order. With $w = 0.9$, $b = 0.2$, $u = 0.55$ and $y = 0.7$ we have $t = 0.6011844863$ and $\partial L/\partial w = -0.0347057345$. Table 1 compares the numerically computed contrasts against (17); the largest deviation is 4.7×10^{-11}.

β	one-sided $\|D - \partial L/\partial\theta\|$	centred $\|D - \partial L/\partial\theta\|$
4.00×10^{-2}	7.968×10^{-2}	5.125×10^{-3}
2.00×10^{-2}	4.096×10^{-2}	1.283×10^{-3}
1.00×10^{-2}	2.076×10^{-2}	3.214×10^{-4}
5.00×10^{-3}	1.045×10^{-2}	8.111×10^{-5}
2.50×10^{-3}	5.245×10^{-3}	2.116×10^{-5}
fitted order	1.00	1.94

Table 2: Contrast error against the exact gradient of the free-phase loss on a three-patch graph with a feedback edge. The centred form used by implementations gains three orders of magnitude at $\beta = 2.5 \times 10^{-3}$.

6.2 Settling, coercion and the role of the prior

The second example is a three-patch graph, p_0 reading two sensory values, p_1 reading x_0, a sensory value and x_2 (a feedback edge), and p_2 reading x_1, with $\lambda = 0.01$. The repair trajectory decreases the energy at every accepted sweep over 2001 sweeps, from 0.22303138 to 0.00396407, and terminates with a projected stationarity residual of 4.297×10^{-13}.

The coercivity bound of Theorem 2 is checked directly: at a random state with $\|x\|_\infty \approx 20$, $E = 209.1931$ while $\frac{1}{2}\sum_i (|x_i| - 1)^2 = 205.0945$, so the bound holds with the expected slack, and it holds identically at $\lambda = 0$. The role of the prior appears in the solution obtained. With $\lambda = 0$ the solve reaches the consistent configuration exactly: $E^\star = 0$, largest prediction error 3.3×10^{-16}. With $\lambda = 0.01$ it reaches a compromise: $E^\star = 0.00396407$ with largest prediction error 1.4×10^{-2}. Both are stationary points corresponding to different objectives. This is the quantitative form of the statement that qualification measures stationarity and not prediction error.

6.3 Order of the contrast

On the same three-patch graph with the feedback edge, the contrast is compared against the true gradient of the free-phase loss, obtained by central differences of the settled free state, which is the implicit-function-theorem gradient and not an approximation to it. Table 2 reports both contrasts. Halving β halves the one-sided error and quarters the centred error; the fitted orders are 1.00 and 1.94. At $\beta = 4 \times 10^{-2}$ the centred contrast is more accurate by a factor of 15.6.

6.4 One operator, and learning that converges

On the same graph and at the same point (x, θ), the state block of the gradient is identical whether or not the parameters are eligible: the measured difference is exactly 0.0, as Theorem 7 requires. A learning solve with a clamped output target decreases its anchored objective monotonically from 0.50822925 to 0.03354746 and terminates with a projected residual of 2.459×10^{-15}. Presenting six successive experiences to the same operator reduces the free-phase cost of the clamped target from 9.036×10^{-3} to 2.649×10^{-4}, while the parameter displacement accumulates steadily. This is characteristic of a system that learns from each experience without discarding what it has already acquired.

6.5 Agreement with the reference implementation

The second script imports the reference library at version 0.80.0 and compares, on the same graph, state and parameters: the energy, the prediction, the error, and the state, weight and bias gradients.

The largest absolute difference over all of them is 1.1×10^{-16}, which is float64 rounding. Driving both implementations from the same initial state with the same budget produces terminal states agreeing to 1.018×10^{-12} and terminal energies agreeing to 2.2×10^{-19}. A settlement through the library's public composition interface qualifies with a stationarity residual of 2.8×10^{-7} after seven accepted sweeps. The mathematics in this paper therefore describes the implementation itself rather than an idealisation of it.

6.6 The evolutionary layer

For the evolutionary example we use scalar meta-parameter φ standing for plasticity, a lifetime of three local updates on a task drawn from the environment, and a metabolic price λ_φ for plasticity. The post-learning fitness is $\hat{J}(\varphi) = -\frac{1}{2}(1-\varphi)^6 - \lambda_\varphi \varphi^2$ in closed form, whose maximum for $\lambda_\varphi = 0.35$ is at $\varphi^\star = 0.3913$; the estimated maximum on a grid of 161 values agrees with the closed form to 7.2×10^{-3}, which is the Monte-Carlo error of the estimate. The optimum being interior and positive is the statement that learning is retained; a value of zero would correspond to a lineage that has eliminated plasticity.

Solving (14) on the grid with mutation strength $\kappa = 0.02$ decreases the population free energy monotonically from 0.379150 to 0.109856, and the population measure approaches the Gibbs form $e^{\hat{J}/\kappa}$ to a total-variation distance of 0.0040. An individual-based simulation with $N = 4000$ individuals over 400 generations, in which each individual performs its own lifetime of updates and is selected on a noisy estimate of its fitness, reaches mean 0.4262 ± 0.1402 against the mean-field prediction 0.4001. The mean-field flow is thus the limit of the finite population, and the residual spread is the finite-population drift that Theorem 12 does not claim to capture at first order.

7 Limits and falsifiable predictions

7.1 Open problems

1. **Uniform hyperbolicity of the fast flow.** Theorem 12 assumes the lifetime dynamics attract uniformly across meta-parameters. The settling theorem, Theorem 3, guarantees convergence for each fixed configuration, but not uniformly with a rate bounded away from zero as the configuration varies. This is the weakest assumption in the argument and the most productive direction for further work.

2. **Non-differentiable meta-parameters.** Part of φ is combinatorial, namely the wiring mask. The fitness is then not differentiable in φ and the averaging theorem needs a stochastic approximation or evolution-strategy form, with weaker convergence conditions.

3. **Estimator noise inside the fitness.** When $\hat{J}$ is estimated from finitely many tasks, selection and estimation noise interact across generations and the covariance terms of Proposition 14 no longer vanish at the nominal stationary point.

4. **Consolidation is not yet inside the unified averaging.** Offline reorganisation at an intermediate time scale is described as a constrained information projection, which is the mechanism that limits catastrophic forgetting, but its scale separation has not been folded into Theorem 12.

5. **Scale.** Every result in this paper is on graphs of a few patches, and the largest systems in the reference line of work are small by the standards of contemporary language models. No result here supports a claim about scale.

7.2 Predictions

The following predictions are stated so that they are falsifiable.

1. **Order of the centred contrast.** Any patch net whose second phase is an equilibrium branch of a C^2 family should show a centred contrast error scaling as β^2 and a one-sided error scaling as β. A measured first-order centred error refutes Theorem 11.

2. **Anchor weight above curvature.** For an anchored learning solve, setting ρ above the local curvature of the data term should remove the oscillatory or divergent parameter trajectories seen at small ρ, and the improvement should track the contraction factor of Theorem 5.

3. **Averaging order.** Doubling the lifetime-to-generation ratio should halve the deviation between the individual-based population and the averaged flow. A measured exponent away from one, for instance a square-root drift-dominated scaling, refutes the hypotheses of Theorem 12.

4. **Weismann boundary.** On a broad or rapidly changing task distribution, the population that inherits only plasticity should reach strictly higher stationary fitness than the population that also inherits learned weights, and on a narrow stable distribution the ordering should reverse.

8 Conclusion

A learning system requires neither a backward pass nor a phase boundary between training and use. It requires an energy, a local repair rule that descends that energy, and a single channel through which the objective enters. We have shown that when the repair is projected gradient descent on the patch energy used by the reference implementation, the descent is coercive, the iterates converge to a single stationary point, inference and learning are one operator restricted to different coordinate sets, and the contrast computed by that operator is second order rather than first. The evolutionary layer is defined on the same structure and is not merely an analogy: its effective fitness is provably the post-learning fitness, its compliance with the formal criteria of biological evolution is proved rather than asserted, and the restriction that acquired weights are not inherited is shown to be advantageous on the task distributions where the distinction matters. The remaining gaps have been stated, and no claim about scaling is made.

A Reproduction

The evidence in §6 is produced by

```
python3 src/verify_eep.py     # Parts A-F, numpy only, Python 3.9+
PYTHONPATH=../cadence/src python3.13 src/cadence_crosscheck.py
```

The first script asserts each claim and exits non-zero if a claim fails. Its parts are: A, analytic gradients against central differences (1.5×10^{-10}); B, settling monotonicity, convergence and coercivity; C, the order of the one-sided and centred contrasts; D, one operator across two coordinate blocks; E, the evolutionary layer; F, the closed-form case of §6.1. The second script imports the reference library and reproduces §6.5. Run logs are archived beside the scripts.

B Structural agreement with observer-patch consensus

The construction of this paper has the shape of a consensus protocol. A finite set of carriers holds local state with a declared interface, disagreement is measured by a scalar functional, accepted local repairs drive that functional strictly down, and the limit is a public normal form that does not depend on the order of repairs. The physics programme that shares this shape derives its consequences from three axioms, a finite carrier with a declared boundary, agreement on overlaps, and no structure beyond the records [19]. The correspondence here is structural and not physical. The learning case adds the second time scale: consensus among patches corresponds to settling and consensus among lineages to selection, and the two are related by the averaging theorem. This parallel is offered as reason to expect the structure to be reusable, and not as evidence for either programme.

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